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Lecture 7: Sinkhorn Divergence, Wasserstein Barycenter

1 Sinkhorn divergence

1.1 From entropic OT to Sinkhorn divergence

Recall that in normal OT, a distribution is always at zero cost to itself, and we can see this by taking the identity coupling. However, this is no longer true for the entropy-regularized OT (EOT). The regularized objective

$$W_\epsilon(\alpha,\beta) = \min_{\pi\in\Pi(\alpha,\beta)} \langle c,\pi\rangle + \epsilon D_{\mathrm{KL}}(\pi\|\alpha\otimes\beta) \tag{1}$$

has two nonnegative competing terms when we let $\alpha=\beta$ and try to minimize. Setting $\langle c,\pi\rangle = 0$ forces the coupling π to live on the diagonal, and setting $D_{\mathrm{KL}}(\pi\|\alpha\otimes\alpha)=0$ forces the independent coupling $\pi=\alpha\otimes\alpha$, which spreads mass off the diagonal. Therefore, unless α degenerates to a single point mass or the cost to zero everywhere, the minimum must be positive. We say that EOT is *biased* in this sense. One consequence we encounter when implementing EOT naïvely in practice is that for the minimization problem $\min_\alpha W^p_{\epsilon,p}(\alpha,\beta)$, the larger ϵ gets, the more we penalize entropy, and as a result the minimizer collapses to a single point mass.

In order to debias EOT, we consider the following objective, which is equivalent to (1) up to an additive constant depending on the marginals:

$$\mathrm{OT}_\epsilon(\alpha,\beta) = \min_{\pi\in\Pi(\alpha,\beta)} \langle C,\pi\rangle - \epsilon H(\pi), \qquad H(\pi) = -\sum_{xy}\pi_{xy}(\log\pi_{xy}-1). \tag{2}$$

The idea is to subtract the nonzero self-costs $\mathrm{OT}_\epsilon(\alpha,\alpha)$ and $\mathrm{OT}_\epsilon(\beta,\beta)$ from (2), which leads us to the definition of the Sinkhorn divergence.

Definition 1. *[RTC17, SZRM18, SBRL18, GPC18] The Sinkhorn divergence between two distributions is given by*

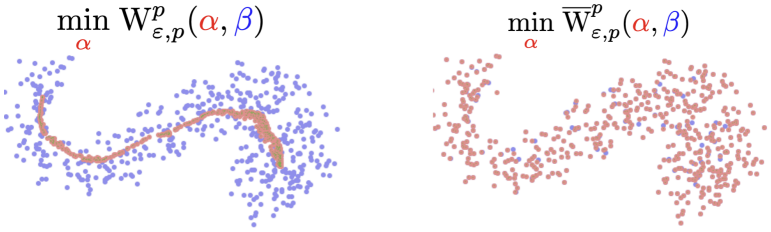
$$S_\epsilon(\alpha,\beta) \coloneqq OT_\epsilon(\alpha,\beta) - \frac{1}{2}OT_\epsilon(\alpha,\alpha) - \frac{1}{2}OT_\epsilon(\beta,\beta) = W_\epsilon(\alpha,\beta) - \frac{1}{2}W_\epsilon(\alpha,\alpha) - \frac{1}{2}W_\epsilon(\beta,\beta).$$

One readily sees that this is equal to zero when $\alpha=\beta$. With reasonable assumptions on the cost function, one can show the following properties of the Sinkhorn divergence.

Proposition 1.
1. *Non-negativity:* $S_\epsilon(\alpha,\beta)\geq 0$.
2. *Symmetry:* $S_\epsilon(\alpha,\beta) = S_\epsilon(\beta,\alpha)$.
3. *Identifiability:* $S_\epsilon(\alpha,\beta)=0 \quad\Leftrightarrow\quad \alpha=\beta$.
4. $S_\epsilon(\alpha,\beta)$ *does NOT satisfy the triangular inequality.*
5. *Metrization of convergence in law:* $\alpha_n \rightharpoonup \alpha \Leftrightarrow S_\epsilon(\alpha_n,\alpha)\to 0$.

Proof. See the proofs of $1,3,5$ in [FSV+19]. An explicit counterexample to 4 is given in [LLM+24]. □

Here's a picture illustrating that the Sinkhorn divergence is really what we want: the entropic OT has the shrinkage effect but by modifying it to Sinkhorn divergence the entropy bias gets removed.



1.2 Sinkhorn divergence as an interpolation

Recall that the version $\mathrm{OT}_\epsilon(\alpha,\beta)$ of EOT interpolates between unregularized $\mathrm{OT}(\alpha,\beta)$ when $\epsilon \to 0$ and the independent coupling $\alpha\otimes\beta$ as $\epsilon\to\infty$. This property of interpolation is also enjoyed by the Sinkhorn divergence. Using the result $\mathrm{OT}_\epsilon(\alpha,\beta) \xrightarrow{\epsilon\to 0} \mathrm{OT}(\alpha,\beta)$ we easily deduce that $S_\epsilon(\alpha,\beta)$ converges to classic OT as $\epsilon\to 0$. On the other hand, when $\epsilon\to\infty$, Sinkhorn divergence connects OT to the maximal mean discrepancy (MMD).

Definition 2. *A Hilbert space $\mathscr{H}$ of functions from a set Ω to $\mathbb{R}$ is a reproducing kernel Hilber space (RKHS) if for each fixed $x\in\mathscr{H}$, the evaluation functional $L_x:\mathscr{H}\to\mathbb{R}, L_x(f):=f(x)$ is continuous. Equivalently (by the Riesz representation theorem), $\mathscr{H}$ is an RKHS if for every $x\in\Omega$, there exists $k_x\in\mathscr{H}$ such that $\langle f,k_x\rangle_{\mathscr{H}}=f(x)$. The reproducing kernel is defined by $k:\Omega\times\Omega\to\mathbb{R}, k(x,y)=\langle k_x,k_y\rangle_{\mathscr{H}}$.*

Conversely, given a symmetric, positive-definite kernel $k:\Omega\times\Omega\to\mathbb{R}$, the Moore-Aronszajn theorem asserts the existence of a unique RKHS $\mathscr{H}$ on Ω for which k is the reproducing kernel.

Definition 3. *[GBR+06] The maximum mean discrepancy (associated with RKHS $\mathscr{H}=\mathscr{H}_k$) is defined by*

$$MMD_k(\alpha,\beta) := \|\mathbb{E}_{x\sim\alpha}[k(x,\cdot)]-\mathbb{E}_{y\sim\beta}[k(y,\cdot)]\|_{\mathscr{H}_k}.$$

The map $\alpha\mapsto\mathbb{E}_{x\sim\alpha}[k(x,\cdot)]$ is called the kernel embedding of α in $\mathscr{H}_k$.

Definitions 2 and 3 provide a way to embed distributions on the original space X into a feature space $\mathscr{H}_k$, and the maximum mean discrepancy between two distributions is the distance between their kernel embeddings in the feature space. The MMD also has the following equivalent characterizations:

- Variational form: $\mathrm{MMD}_k(\alpha,\beta)=\sup_{\|f\|_{\mathscr{H}_k}\le 1}\left(\mathbb{E}_{x\sim\alpha}[f(x)]-\mathbb{E}_{y\sim\beta}[f(y)]\right)$.
- Expectation form: $\mathrm{MMD}_k^2(\alpha,\beta)=\mathbb{E}_{x,x'\sim\alpha}[k(x,x')]+\mathbb{E}_{y,y'\sim\beta}[k(y,y')]-2\mathbb{E}_{x\sim\alpha,y\sim\beta}[k(x,y)]$.

One observation we can make here is that the variational form of MMD is analogous to the dual formulation of 1-Wasserstein distance (we call these integral probability metrics), while the expectation form is analogous to the formulation of the Sinkhorn divergence (two self-terms, one cross term, and similar coefficients). Here are some useful properties of MMD [GBR+06]:

- For characteristic kernels, one has $\mathrm{MMD}_k(\alpha,\beta)=0\Leftrightarrow\alpha=\beta$.
- The computation of MMD only involves taking expectations and does not require optimization: this implies a computational complexity of $O(n^2)$.
- MMD can be estimated using sample averages and has statistical complexity $O(n^{-1/2})$.

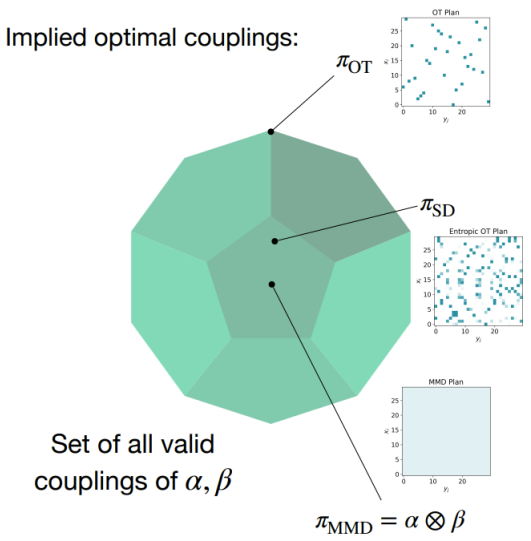
We have previously remarked that the Sinkhorn divergence interpolates between classic OT and MMD. The following result is a rigorous formulation of this idea.

Proposition 2. *[RTC17] Let C denote the cost function associated with the OT problem. Then one has*

$$OT(\alpha,\beta)\xleftarrow{0\leftarrow\epsilon} S_\epsilon(\alpha,\beta)\xrightarrow{\epsilon\to\infty}\frac{1}{2}MMD^2_{-C}(\alpha,\beta).$$

(The kernel $-C$ is okay here because we are integrating it against $\alpha-\beta$, which has total mass zero).

To offer a bit of geometric intuition, recall that for the given two marginal distributions α,β, the set of all admissible couplings $\Pi(\alpha,\beta)$ forms a convex polytope. The solutions to classic OT correspond to the corners of this polytope - these hard matchings lead to sparse couplings. As we increase the regularization parameter ϵ from 0, the solutions move to the interior of the polytope, giving softer, more dense couplings. Finally, as $\epsilon\to\infty$, the entropy term dominates and thus the optimal coupling converges to the minimum-entropy one, namely the independent coupling $\alpha\otimes\beta$. In view of Proposition 2, this can also be referred to as the MMD coupling because the squared MMD does not search over couplings but instead has a cross term that takes expectation over $\alpha\otimes\beta$.



1.3 Complexity of Sinkhorn divergence

In this subsection, we shall see that similar to EOT, the Sinkhorn divergence is an improvement upon classic OT in terms of both computational and statistical complexity. In previous lectures, we have discussed that classical OT, fomulated as an LP, has computational complexity $O((n^2)^3) = O(n^6)$ via the traditional interior point methods. This can be improved to $O(n^3 \log n)$ specifically for the OT problem ([KS95]), but it still pales in comparison with Sinkhorn divergence.

Proposition 3. *[DGK18] The computational complexity of Sinkhorn divergence is $\tilde{O}(n^2/\epsilon^2)$ to obtain an ϵ-accurate solution.*

Proof. This a direct consequence of the computational complexity of the Sinkhorn algorithm for EOT (add reference), since we can compute the three terms of $S_\epsilon(\alpha, \beta)$ in parallel. Note also that self-terms are empirically easier to compute because the marginals are symmetric and the optimal coupling is not too far from the identity coupling. □

Another quantity we are interested in is the statistical complexity. (Here we briefly change notation to use N for the sample size and d for dimension.) In practice, when the only way we can access a distribution is to draw N i.i.d. samples from it, this naturally raises the question of how fast a sample statistic converges to the truth. For instance, in classic OT, if we would like to estimate the distance $W_p(\alpha, \beta)$ between two distributions by the distance $W_p(\hat{\alpha}_N, \hat{\beta}_N)$ between the empirical distributions, there is the classic curse of dimensionality:

Proposition 4. *[FG14] With mild assumptions, for $d > 2p$ one has*

$$\mathbb{E}\left[\left|W_p(\hat{\alpha}_N, \hat{\beta}_N) - W_p(\alpha, \beta)\right|\right] = O(N^{-1/d}).$$

However, for EOT (and consequently for Sinkhorn divergence), we have the following result:

Proposition 5. *[GCB+19] With mild assumptions, one has*

$$\mathbb{E}\left[\left|S_\epsilon(\alpha, \beta) - S_\epsilon(\hat{\alpha}_N, \hat{\beta}_N)\right|\right] \asymp \mathbb{E}\left[\left|W_{p,\epsilon}(\hat{\alpha}_N, \hat{\beta}_N) - W_{p,\epsilon}(\alpha, \beta)\right|\right] = O(\epsilon^{-d/2} N^{-1/2}) \overset{\epsilon \to \infty}{\longrightarrow} O(N^{-1/2}).$$

We note here that when $\epsilon \to \infty$, the rate is $N^{-1/2}$, which is independent of d and the best possible in most classical statistical problems. We have now seen that the Sinkhorn divergence outperforms classic OT in both computational and statistical complexity. To connect to the theme of interpolation in the previous subsection, we include the following remark on the complexity of MMD for completeness.

Remark 1. *[GBR+06] The computational complexity of MMD is $O(n^2)$, and the statistical complexity of MMD is $O(N^{-1/2})$ (which is dimension-free). Below is a table comparing MMD, Sinkhorn and the classical OT.*

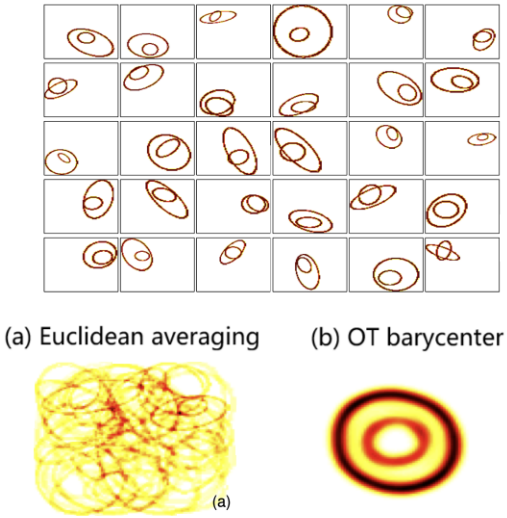
	Classic OT	Sinkhorn Divergence	MMD
Computational Complexity	$\mathcal{O}(n^3 \log n)$	$\mathcal{O}(n^2/\epsilon^2)$	n^2
Statistical Complexity	$\mathcal{O}(n^{-\frac{1}{d}})$	$\mathcal{O}(\epsilon^{-\frac{d}{2}} n^{-\frac{1}{2}})$	$\mathcal{O}(n^{-\frac{1}{2}})$
Coupling	Hard/sparse coupling	Soft/dense coupling	Naive (independent) coupling
Geometry	Via cost	Via cost	Via kernel

2 Wasserstein Barycenters

In machine learning, there is often a need to aggregate, combine, and summarize data. For simple data points, this can be done via sample statistics, clustering, nearest neighbor, etc. These methods have geometric interpretations - for instance, the sample mean can be viewed as the center of the data points in Euclidean geometry, i.e.,

$$x^* = \arg\min_x \sum_{i=1}^K \|x - x_i\| \quad \Rightarrow \quad x^* = \frac{1}{K} \sum_{i=1}^K x_i.$$

A natural first attempt is to average the data in the Euclidean sense, but this leads to the bottom-left result—a noisy, uninformative mixture. In contrast, the bottom-right figure illustrates the outcome we would intuitively expect: a clean and representative barycenter. In the following section, we introduce how optimal transport provides a principled way to obtain such desirable averages.



It turns out that there are also compelling motivations to in some sense find the center of our data when they are probability distributions instead of points. This approach is particularly useful for image color grading ([BPC16]), dataset homogenization ([KELB22]), sensor data aggregation ([SCSJ17]). Finding the the geometric center in the space of distributions allows us to: 1) combine multiple domains or clients (federated learning); 2) compress large datasets into representative averages; 3) morph between styles, textures, or modalities. We would like to extend the notion of taking the average to the space of distributions without ignoring its underlying geometry, which leads us to the definition of the Wasserstein barycenter:

Definition 4. *Given measures $\{\mu_i\}$ weights $\{\lambda_i\}$, the W_p-barycenter is given by*

$$\nu^* = \arg\min_{\nu} \sum_{i=1}^{K} \lambda_i W_p^p(\mu_i, \nu).$$

Note that if all the λ_i's are equal, then ν^* is equidistant from the μ_i's; if we increase a specific λ_i, then $W_p^p(\mu_i, \nu)$ weighs more and draws ν^* closer to μ_i. The following are some useful observations and properties of the Wasserstein barycenter:

- The Wasserstein barycenter value is convex in the weights $\{\lambda_i\}$.
- The Wasserstein barycenter is unique under some mild assumptions (e.g., for W_2 in $\mathbb{R}^d$, if at least one of the input measures is absolutely continuous w.r.t. the Lebesgue measure, then the barycenter is unique).
- If $p = 2, \mu_i = \delta_{x_i}, \lambda_i = 1/K$, then the barycenter recovers the Euclidean mean $\nu^* = \delta_{\text{mean}(x_i)}$. If $p = 1, \mu_i = \delta_{x_i}, \lambda_i = 1/K$, then the barycenter recovers the median $\nu^* = \delta_{\text{median}(x_i)}$. This is another example of how the OT geometry recovers Euclidean geomtry in corner cases.
- For $p = 2$, the mean of the barycenters is equal to the barycenter of the means, i.e.,

$$\int_X x d\nu^*(x) = \sum_i \lambda_i \int_X x d\mu_i(x).$$

- For Gaussian distributions, the W_2-barycenter is also Gaussian, but the geometry is nontrivial in the covariances.

Proposition 6. *Let $\mu_i = \mathcal{N}(m_i, \Sigma_i)$. Then the barycenter is a Gaussian $\mathcal{N}(m^*, \Sigma^*)$ with $m^* = \sum_{i=1} \lambda_i m_i$, and Σ^* the solution to the nonlinear matrix equation:*

$$\Sigma^* = \arg\min_{\Sigma} \sum_i \lambda_i \mathscr{B}(\Sigma_i, \Sigma),$$

where $\mathscr{B}$ is the Bures metric.

Distributions live in a curved space whose geometry is given by the Wasserstein metric. The Wasserstein barycenter is then the center of mass with respect to this particular geometric structure: it minimizes the weighted average transport cost instead of naïvely taking the pointwise average density. We note here that computing the Wasserstein barycenter between two measures yields the McCann interpolant.

The OT barycenter in the discrete case has the following LP formulation.

$$\min_{\substack{a \in \Delta_n \\ (\pi_s \in \mathbb{R}^{n \times n_s})_s}} \left\{ \sum_{i=1}^{K} \lambda_s \langle \pi_s, C_s \rangle : \forall s, \pi_s \mathbf{1}_{n_s} = a, \pi_s^T \mathbf{1}_n = b_s \right\} \tag{3}$$

For each s, the matrix π_s is a coupling between the source distribution a and target distribution b_s. The LP problem minimizes the weighted average transport cost. However, this LP is computationally heavy and impractical to solve exactly in most ML cases. One way to remedy this is that we perform first order methods instead of interior-point solves, e.g., subgradient ascent on the dual potentials.

To simply computation, again we can regularize each individual OT with entropy and make the OT barycenter problem smooth and strictly convex:

$$a^* = \arg\min_{a} \sum_{s=1}^{K} \lambda_s \text{OT}_\epsilon(a, \beta_s)$$

Keeping the same notations as before, a Sinkhorn-type iteration is

1. $v_s^{(t)} \leftarrow a \oslash (K_s u_s^{(t)})$ in parallel

2. $a^{(t)} \leftarrow \prod_s (K_s v_s^{(t)})^{\lambda_s}$ then normalize

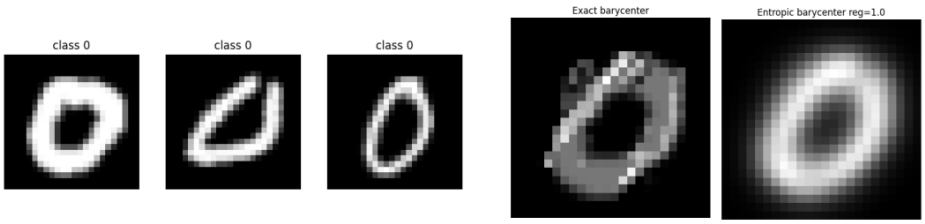
3. $u_s^{(t)} \leftarrow b_s \oslash (K_s^T v_s^{(t)})$ in parallel

Similar to EOT, this regularized version will give us a unique solution that is smooth, more dense, and with faster convergence guarantees.

We conclude with a remark comparing classical barycenters and entropic barycenters.

- Classical barycenters preserve exact, sharp features but are computationally heavy and ofter undesirable in practice.
- Entropic barycenters are approximate and more blurred, but in practice they are much faster to compute and scalable.

In the figure below, we compute Wasserstein barycenters of digit images from the MNIST dataset. The classical barycenter (left) captures crisp structural details of the digits but is expensive to compute. In contrast, the entropic barycenter (right) yields a smoother and more diffuse representation, reflecting the regularization effect. While less sharp, the entropic version is computationally efficient and scales better, making it the practical choice in many applications.



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